

Episodex: Continuous Context Management for Generalist Agents through Episodic and Conceptual Memory

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Abstract

We introduce Episodex, a cognitively inspired Context Management Strategy (CMS) designed to address the limitations of transformer-based language models in long-horizon, interactive environments. Episodex structures an agent’s experience into coherent episodes of belief states and clusters conceptual knowledge for symbolic reuse, enabling a continuous context window that scales beyond the bounds of the model’s attention span. The design of Episodex is grounded in empirical observations from experiments conducted in the ALFWorld environment using a meta-agent architecture inspired by Global Workspace Theory (GWT). Our implementation uses a system of 12 modular cognitive agents coordinated through an LLM-driven global workspace. We find that this architecture enables interpretable, modular reasoning and supports dynamic memory integration, leading to improved efficiency and robustness in task completion. Episodex thus provides a practical foundation for managing context in generalist agents operating under token limitations.

1 Introduction

Language agents built on large language models (LLMs) are increasingly being deployed in complex, multi-step environments where they must reason over extended sequences of observations and interactions. A persistent challenge in such settings is the limitation of the transformer’s context window: as the number of tokens in an agent’s interaction history increases, relevant information can be truncated or lost, leading to degraded performance.

To address this, we propose **Episodex**, a Context Management Strategy (CMS) that segments an agent’s interaction history into semantically meaningful episodes, stores these episodes in long-term memory, and enables retrieval of relevant prior experiences during task execution. Episodex draws inspiration from cognitive science—particularly

Global Workspace Theory (GWT), dual-system reasoning (System 1/System 2), and the structure of human episodic memory—and aims to operationalize these insights in computational systems.

Episodex builds on and synthesizes advances in multi-agent frameworks like AutoGen (Wu et al., 2023), memory-augmented agents like CoALA (Sumers et al., 2024), symbolic reasoning strategies like Chain-of-Thought prompting (Wei et al., 2022), and control architectures like meta-agents (Hu et al., 2024). In particular, Episodex integrates these directions under a unified context strategy inspired by ReAct (Yao et al., 2023), where agents reason and act cyclically.

2 Related Work

A wide range of research efforts have explored context management, memory augmentation, and cognitive frameworks in large language model (LLM) agents.

Chain-of-Thought Prompting. Chain-of-thought (CoT) prompting (Wei et al., 2022) improves problem-solving in LLMs by encouraging step-by-step reasoning. It highlights the benefits of generating intermediate reasoning steps rather than end-to-end answers.

ReAct. ReAct (Yao et al., 2023) proposes an interleaved strategy of reasoning and acting, where LLM agents reflect on their environment, take actions, and update plans iteratively. This method laid the foundation for reasoning-augmented interactive agents.

Reflexion. Reflexion (Shinn et al., 2023) extends ReAct by incorporating self-critiquing loops into the agent workflow. These loops allow agents to learn from mistakes and adapt strategies across tasks through verbal reinforcement learning.

AutoGen. AutoGen (Wu et al., 2023) introduces a framework for multi-agent communication via

natural language. It facilitates modular collaboration by treating LLMs as cooperative conversational components without requiring persistent memory structures.

CoALA. CoALA (Sumers et al., 2024) focuses on modular LLM agents equipped with memory systems and external tool use. It formalizes the use of long-term and short-term memory in interactive language agents and emphasizes structured memory interfacing.

EM-LLM. EM-LLM (Fountas et al., 2024) organizes episodic memory by identifying event boundaries using Bayesian surprise and graph-theoretic segmentation. It then clusters these episodes to form compact memory representations.

InfLLM. InfLLM (Xiao et al., 2024) employs fixed-size input segmentation and k-nearest neighbor retrieval to reintroduce relevant context. It emphasizes fast retrieval using similarity-based heuristics.

Conscious Turing Machines (CTM). CTM (Blum and Blum, 2022) proposes a computational model of consciousness based on Global Workspace Theory. It divides memory and processing into modular components coordinated by a central broadcasting mechanism.

Meta-Agent Systems. Meta-agent orchestration (Hu et al., 2024) explores top-level LLMs directing the actions of specialized sub-agents. These systems rely on agent transition graphs and centralized decision-making to coordinate complex reasoning workflows.

Together, these works provide the theoretical and technical foundation for designing modular, memory-aware, and cognitively motivated language agents.

3 Episodex: A Context Management Strategy

3.1 Motivation and Design Principles

Episodex is a cognitively inspired Context Management Strategy (CMS) developed to address the growing need for long-term, scalable reasoning in language agents operating under context length limitations. It structures experience into semantically coherent episodes and abstracts symbolic knowledge into conceptual clusters, enabling efficient reuse and adaptive planning.

While Episodex emerged from our empirical experimentation with a multi-agent architecture in ALFWorld, it is not tied to that implementation. Rather, it generalizes the insights gained from those experiments into a standalone strategy that can be deployed independently across LLM-based agents. For instance, while dynamic segmentation and retrieval-augmented generation (RAG) are part of Episodex’s design, they were not necessary to implement in ALFWorld due to its naturally segmented tasks and high performance under random retrieval. Nonetheless, these components remain essential to generalizing Episodex to open-ended or continuous environments.

Episodex is guided by the following principles:

- **Memory Efficiency:** Maintain a continuous stream of task-relevant context without overwhelming the transformer’s fixed-length window.
- **Abstraction and Generalization:** Organize experience into structured episodes and abstract symbolic knowledge for reuse across tasks.
- **Interpretability:** Ensure that the memory structure remains modular and human-interpretable for inspection and adaptation.

By separating reasoning into semantically coherent episodes and representing learned insights as symbolic concepts, Episodex supports a continuous context window that preserves semantic continuity.

While Episodex has only been partially implemented within our meta-agent framework, it stands as a general model-agnostic strategy that can support a variety of architectures and domains.

3.2 Memory Taxonomy

Implicit Long-Term Memory This resides in the parameters of the LLM itself. It encodes knowledge learned during pretraining, including language structure, factual associations, and heuristics. While powerful, this memory is inaccessible for modification during inference and cannot adapt to novel situations without retraining.

Working and Short-Term Memory Short-term memory corresponds to the current context window available to the transformer model. In contrast, **working memory** refers to the internal activations of the model as it processes this context and generates responses. These activations form the latent

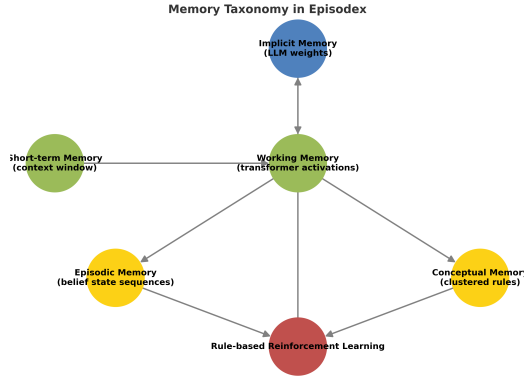


Figure 1: Memory Taxonomy in Episodex. Colors distinguish implicit, working/short-term, explicit, and bridging memory mechanisms.

state of reasoning and can be thought of as dynamic buffers for manipulating information in the service of task goals.

This interpretation is supported by recent work from Anthropic on the biology of large language models (Lindsey et al., 2025), which found that internal transformer activations exhibit memory-like behavior analogous to working memory in biological systems. These activations retain relevant features and intermediate computations across time steps, suggesting that transformer layers act as transient yet cognitively structured memory stores.

In Episodex, this distinction enables architectural clarity: short-term memory contains what the model sees, while working memory encompasses what the model thinks with.

Explicit Long-Term Memory Episodex proposes a structured, external memory designed to complement the LLM’s parametric knowledge. This includes:

- **Episodic Memory:** Temporally ordered sequences of belief states that capture how the agent experienced its environment.
- **Conceptual Memory:** Symbolic abstractions derived from clustering across successful belief trajectories. These represent generalized knowledge or rules that can be applied across tasks.

Concepts are learned dynamically during each episode as the belief state is updated in response to environmental feedback. Clustering and consolidation of concepts occur after the episode concludes—analogueous to memory consolidation during sleep. This process compresses and organizes

learned abstractions for efficient reuse in future tasks.

This memory is structured, interpretable, and selectively retrieved to support the current episode. Episodex aligns with cognitive frameworks such as Conscious Turing Machines (Blum and Blum, 2022), which emphasize the centrality of an active workspace broadcasting to a modular set of memory systems.

3.3 Bridging Memory Systems with Rule-Based RL

While explicit memory operates outside the transformer, our experimental observations suggest that, ultimately, they must interface effectively with the transformer’s implicit parametric knowledge. Despite successful learning and retrieval of useful conceptual abstractions, the meta-agent frequently failed due to persistent reliance on strong implicit priors and mismatches with real-world mechanics (e.g., unrealistic microwave behaviors in ALF-World). These consistent failure modes highlighted the necessity for adaptive conceptual models capable of real-time adjustments. In response, Episodex proposes Online Rule-based Reinforcement Learning (RL) as a solution to dynamically bridge explicit conceptual memory and implicit transformer knowledge. By providing symbolic structures derived from conceptual memory directly as model inputs and using immediate feedback through reward signals, this method continuously refines and aligns conceptual knowledge to overcome real-world mismatches and improve agent performance.

3.4 System Architecture

A complete Episodex system consists of the following interacting components:

- **Short-Term Memory:** The context window processed at each step.
- **Working Memory:** Transformer internal representations that evolve during token generation, consistent with activation-based memory structures (Lindsey et al., 2025).
- **Episodic Memory Base:** A growing database of episodes segmented by belief state transitions.
- **Conceptual Knowledge Base:** Symbolic rules or clusters abstracted from episodic traces that grows sub-linearly.

Episodes are formed through belief state updates. At the end of each episode, the full sequence is archived and relevant abstractions are extracted. During future tasks, relevant episodes and concepts are retrieved to inform planning and reasoning.

4 Comparative Analysis with Existing Work

Episodex introduces a structured, cognitively inspired context management strategy (CMS) for large language model (LLM) agents, informed by empirical findings from our implementation using evolving belief states and conceptual memory clustering. Here we compare the implemented aspects of Episodex—specifically episodic belief state structures, conceptual memory clustering, and explicit memory taxonomy—to related work.

EM-LLM. The EM-LLM model (Fountas et al., 2024) organizes context into coherent episodic events using Bayesian surprise and graph-theoretic methods. Episodex similarly structures context into coherent episodes, but utilizes sequences of evolving belief states explicitly generated by an agent’s internal reasoning processes rather than Bayesian surprise or event graph refinements. EM-LLM employs semantic clustering for organization, closely paralleling Episodex’s conceptual memory clustering, although Episodex uniquely emphasizes concept clustering derived from successful outcomes and contrastive learning during task execution.

InfLLM. InfLLM (Xiao et al., 2024) segments contexts into fixed-size units and employs k-nearest neighbor (k-NN) lookups for memory retrieval. While InfLLM’s segmentation approach differs from Episodex’s belief-state-driven episodic structuring, both systems utilize semantic similarity retrieval. However, Episodex explicitly maintains conceptual memory derived from structured belief states and task-specific abstraction, potentially improving interpretability and cross-task generalization relative to InfLLM’s simpler, fixed-segment retrieval mechanism.

Reflexion. Reflexion (Shinn et al., 2023) employs reflective loops to iteratively self-improve based on task outcomes. Episodex similarly leverages post-task abstraction of successful actions into generalized symbolic knowledge but places additional emphasis on structuring these insights within a formal conceptual memory base, thereby pro-

moting explicit and reusable knowledge structures beyond simple reflective logs.

Conscious Turing Machines (CTM). CTM (Blum and Blum, 2022) conceptualizes consciousness computationally via a global workspace broadcasting information across modular subsystems. Episodex implements a similar global workspace-like structure through structured episodic and conceptual memories but explicitly categorizes memory into short-term, working, and long-term stores. This explicit memory taxonomy enhances both the interpretability and modularity of memory management in Episodex relative to CTM’s broader conceptual approach.

In conclusion, Episodex aligns with and extends existing frameworks by uniquely structuring memory around evolving belief states, explicit conceptual clustering, and a clearly defined memory taxonomy, offering enhanced interpretability and modularity in managing LLM agent contexts.

5 Experimental Framework: Meta-Agent Implementation

5.1 Meta-Agent Design

To test key elements of Episodex, we implemented components within a modular cognitive agent inspired by Global Workspace Theory. This agent operates using a transition graph of roles (e.g., planning, memory, learning), each implemented as prompt-driven modules.

We present a generalist agent architecture inspired by the Global Workspace Theory (GWT) of consciousness, aiming to unify specialized large language model (LLM) capabilities into a single functional "mind." The underlying LLM acts as an unconscious System 1—handling intuitive, low-level processing—while a multi-agent architecture implements a deliberative System 2, responsible for goal-directed reasoning and memory coordination. We aim to simulate a form of functional consciousness—the dynamic and context-sensitive coordination of cognitive subsystems—not sentience. We do not claim to replicate human subjective experience; instead, we operationalize consciousness as an emergent property of structured interaction with an environment.

Our architecture, evaluated in the ALFWorld environment (Shridhar et al., 2021), comprises 12 specialized agents responsible for perception, memory management, learning, planning, and reflection.

These agents communicate via a centralized global workspace, which serves as the system’s attentional bottleneck. Notably, the LLM orchestrates this agentic system by selecting the next agent to invoke at each step—subject to a directed graph of allowed transitions. This results in an interpretable, mechanistic chain-of-thought process, where the reasoning trace can be decomposed into a sequence of modular agent activations.

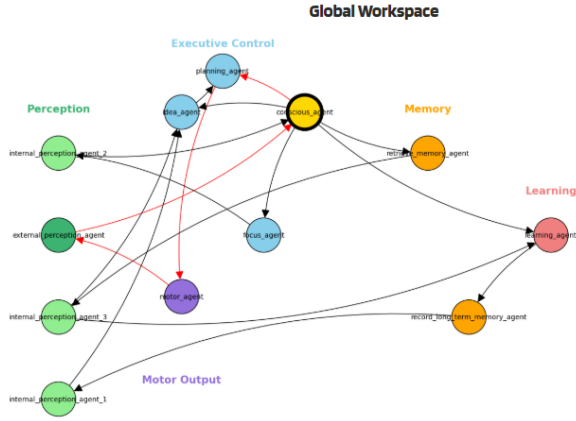


Figure 2: Agent transition architecture detailing the global workspace communication and allowed transitions between specialized agents.

This structure enforces coherent cognitive trajectories—progressing through planning, perception, reflection, memory access, learning, and ideation—while allowing the LLM to control agent execution adaptively within interpretable constraints. Unconscious background subprocesses, such as K-means clustering for long-term memory organization at the start of each task and ongoing memory management, support the operation of the global workspace.

Our model extends recent advances in LLM-based agentic reasoning, including ReAct (Yao et al., 2023), Reflexion (Shinn et al., 2023), and AutoGen (Wu et al., 2023). While ReAct interleaves reasoning and acting and AutoGen introduces modular helper agents, our framework offers a more unified and cognitively inspired design grounded in systems neuroscience and GWT theory. Evaluated on the ALFWorld benchmark, our system achieves a success rate of 87.5% (out of 139 games). These results suggest modeling cognition as a structured, functionally conscious system—driven by interpretable transitions and centralized attention—can lead to more robust and generalizable reasoning in interactive environments.

System 1 is realized as a group-chat orchestrator—typically an LLM such as GPT-4o—that determines which agent to activate and interprets outputs. System 2 corresponds to the structured transition graph and modular agent architecture. This mirrors meta-agent designs explored in Hu et al. (2024), which highlight the utility of centralized control in coordinating multi-agent behavior.

5.2 Episodic and Conceptual Memory Implementation

Within this system, we implemented the following:

- **Episodic Memory:** Each episode is formed by capturing belief states generated by the `conscious_agent` after each ALFWorld action. These are stored as textual traces.
- **Retrieval Mechanism:** For every new task, a fixed number of past episodes are randomly selected and injected into the context window.
- **Conceptual Memory:** After each task, symbolic abstractions are clustered from successful belief sequences. These concepts are stored and retrieved in full alongside episodic memory.

We emphasize that this is a partial implementation. There is no dynamic segmentation, no relevance-based retrieval, and no runtime compression. Nevertheless, this architecture allowed us to investigate how modular memory structures influence agent performance.

5.3 Evaluation Environment: ALFWorld

To empirically explore and refine Episodex, we evaluated its core mechanisms within the ALFWorld benchmark (Shridhar et al., 2021). ALFWorld is a simulation environment that combines embodied task completion with natural language interaction. Agents operate in a 3D household environment via a text interface, issuing commands like “open the fridge” or “put the apple in the microwave.”

Each task consists of a goal (e.g., “heat the apple”), requiring the agent to perform multi-step interactions, reason about object affordances, and track environment state. The environment returns structured observations in response to each command, including visible objects and feedback on action success.

This setting is particularly suitable for testing context management, as tasks unfold over extended sequences of actions with delayed feedback. It also enables measurement of planning quality, memory integration, and symbolic generalization.

6 Experimental Observations

6.1 Setup

We evaluated the meta-agent in ALFWorld, a text-based interactive environment. Tasks required agents to manipulate objects via textual commands and interpret structured responses.

6.2 Quantitative and Qualitative Findings

- **Efficiency Gains:** Increasing the number of retrieved episodic memories enhanced efficiency by reducing the average number of actions per successful task from 17 to 10, though this increase did not significantly affect overall success rates.

Max Retrieved Episodes	Avg. Actions
0	17.0
5	14.8
10	12.6
15	11.2
20	10.0

Table 1: Efficiency gains through episodic memory retrieval over 20 games. As the number of retrieved episodes increases, the average number of actions required per successful task decreases.

- **Clustered Conceptual Memory Retrieval:** Retrieval of clustered conceptual memory significantly improved task performance, increasing the success rate from 60% to 82.5%.
- **Enhanced Planning Prompts:** Further improvement by embedding environmental affordances into planning prompts boosted the success rate from 82.5% to 87.5%.
- **Cognitive Strategy:** The group-chat manager prioritized short, low-cognitive-cost feedback loops (*autopilot*), engaging deeper cognitive modules only when necessary.
- **Structured Reasoning:** Effective transitions between specialized agents (System 2) allowed complex tasks to be successfully decomposed and solved through structured, modular reasoning.



Figure 3: Normalized transition probabilities between cognitive agents during a representative ALFWorld run over 63 games.



Figure 4: Transition frequency heatmap between cognitive agents during a representative ALFWorld run over 63 games.

7 Future Directions

To improve our current Episodic implementation, we identify several directions:

- **Dynamic Episode Segmentation:** Use novelty or surprise signals to segment episodes more intelligently.
- **Retrieval-Augmented Generation (RAG):** Implement learned retrieval over episodic memory to increase relevance.
- **Compression and Prioritization:** Prioritize or compress past memory to prevent information overload.

8 Conclusion

Episodex proposes a general context management framework inspired by cognitive science. It segments experience into episodes and abstracts symbolic knowledge for retrieval-based reasoning. Though only partially implemented, our prototype within a meta-agent framework demonstrates the potential of structured memory and modular control in mitigating context window limitations and enabling scalable reasoning.

Limitations

Our current implementation is a proof of concept. It uses fixed retrieval, lacks dynamic segmentation, and was tested only in text-based environments. We do not yet evaluate generalization to multimodal settings or continuous control tasks.

Ethics Statement

This work promotes interpretable memory systems and does not involve human subjects or personal data.

References

- Lenore Blum and Manuel Blum. 2022. A theory of consciousness from a theoretical computer science perspective: Insights from the conscious turing machine. *Proceedings of the National Academy of Sciences (PNAS)*, 119(21):e2115934119.
- Zannis Fountas, Albert Webson, Xuezhi Han, Luke Zettlemoyer, and Tatsunori B. Hashimoto. 2024. [Episodic memory in large language models](#).
- Shengran Hu, Cong Lu, and Jeff Clune. 2024. Automated design of agentic systems. *arXiv preprint arXiv:2408.08435*.
- Reed Lindsey, Noemi Derzsi, Ethan Perez, and Jared Kaplan. 2025. On the biology of a large language model. *Anthropic Technical Report*. <https://www.anthropic.com/research/biology-of-a-large-language-model>.
- Noah Shinn, Andy Zou, and Piero Molino. 2023. [Reflexion: Language agents with verbal reinforcement learning](#). *arXiv preprint arXiv:2303.11366*.
- Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, and Dieter Fox. 2021. [Alfworld: Aligning text and embodied environments for interactive learning](#). In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. 2024. Cognitive architectures for language agents. *Transactions on Machine Learning Research (TMLR)*.

- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*.
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed H. Awadallah, Ryen W. White, Doug Burger, and Chi Wang. 2023. Autogen: Enabling next-gen llm applications via multi-agent conversations. *arXiv preprint arXiv:2308.08155*.
- Ziru Xiao, Luke Zettlemoyer, and Tatsunori B. Hashimoto. 2024. [Infinite context windows for language models](#).
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. React: Synergizing reasoning and acting in language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.